INCH-POUND

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30 June 1989
SUPERSEDING
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4 June 1981

MILITARY SPECIFICATION

CASE, TRANSFER, HUMAN REMAINS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of transfer case for human remains.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 9930

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SPECIFICATIONS

FEDERAL

U-T-30	-	Talcum Powder
QQ-A-250/1	-	Aluminum 1100, Plate and Sheet
QQ-A-250/2	-	Aluminum Alloy 3003, Plate and Sheet
TT-E-529	-	Enamel, Alkyd, Semigloss
MMM-A-121	-	Adhesive, Bonding Vulcanized Synthetic Rubber to Steel
MMM-A-1617	-	Adhesive, Rubber Base, General Purpose

MILITARY

MIL-R-3065	-	Rubber, Fabricated Products
MIL-A-8625	-	Anodic Coatings, for Aluminum and Aluminum Alloys
* MIL-V-27166	-	Valve, Pressure Equalizing, Gaseous Products
MIL-T-43566	-	Tape, Textile, Cotton, General Purpose, Natural or in Colors

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-417	-	Classification System and Tests for Solid Elastomeric Materials
MS-20470	-	Rivet, Solid-Universal Head, Aluminum Alloy and Titanium Columbian Alloy

* (Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWING

US ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

5-13-2106	-	Case, Transfer; Human Remains

(Copies of drawings are available from the U.S. Army Natick Research, Development, and Engineering Center, ATTN: STRNC-EMSS, Natick, MA 01760-5014.)

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* 2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Welding Qualifications of the American Society of Mechanical Engineers

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

* AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 633 - Electrodeposited Coatings of Zinc on Iron and Steel
D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

* (Non-Government standards and other publications are normally available from organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3), in accordance with 4.3.

3.2 Alternate components. Components offered as equivalent to components specified hereinafter and on the referenced drawing as a specific manufacturer's part number or equal shall be functionally equal to the manufacturer's part number identified and of equal or better quality. The incorporation and inclusion of such a component in the design of the specified end product shall not require modification or change to any other specified component, and shall not reduce ease of maintenance to it or any other components, unless such modification or change is specifically approved by the contracting officer. Prior to manufacture of the first article or if none is required, prior to commencing production, the contractor shall submit for the contracting officers' approval, a list identifying each proposed alternate component, together with proof that each listed component complies with

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requirements specified herein. The contracting officer, at his option, may require a physical sample of the proposed substitution. Approval of the submitted listing, together with necessary supporting data, authorizes the commencement of the first article or production manufacture, as applicable, and does not relieve the contractor of responsibility that these components perform in accordance with specified requirements when incorporated into the end product.

* 3.3 Materials and components. The materials and components shall be as specified herein and as shown on the applicable drawing. Materials and components not definitely specified shall be of the quality normally used by the manufacturer provided the completed item complies with all the provisions of this specification. It is encouraged that recycled material be used when practical as long as it meets the requirements of this document.

* 3.4 Design and construction. Design and construction for the case shall be as specified herein and as shown on Drawing 5-13-2106. Runners shall be of one piece construction. Rivets shall conform to AD6-7 of MS 20470. Case seams prior to anodizing shall be closed by continuous welding on the inside (where accessible) and the outside. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion.

* 3.4.1 Aluminum. Unless otherwise specified, the aluminum plate and sheet shall conform to 1100-H14 in accordance with QQ-A-250/1, or 3003-H14 in accordance with QQ-A-250/2.

* 3.4.2 D-rings and tapes. D-rings shall be 1-1/4 inches wide and 1-1/8 inches high and shall be constructed from 0.2437 inch diameter steel wire. D-rings shall be zinc plated in accordance with type I or II, (with a minimum thickness of 13 micrometers) of ASTM B 633 or shall have a black enamel finish conforming to type II of TT-E-529. The bottom unit shall have six D-rings secured to the inside by aluminum alloy clips, riveted as shown on the drawing. Three tapes, 1 inch wide by 9 feet long shall conform to type I, class 1 of MIL-T-43566. Each tape length end shall be threaded through one D-ring on one side of the bottom of the case, and then, through the other D-ring on the opposite side of the case. The loose ends of the tape shall be tied together.

* 3.4.3 Handles. The case shall have ten spring loaded carrying handles. The handle grips and mountings shall be aluminum. The bails and springs shall be corrosion-resisting steel. The opening of the grip shall be not less than 4 inches. The handles shall be riveted after the case has been anodized.

* 3.4.4 Bottom, top and runners. The bottom, top and runners of the case shall be fabricated from 0.090 inch aluminum sheet as specified in 3.4.1. Runners shall be spot welded to the case. The spot weld spacing shall not be greater than 3-1/2 inches between centers.

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3.4.5 Gasket. The gasket material shall conform to SC 415A, B, F or RN 315A of MIL-STD-417 and MIL-R-3065 except that the Durometer hardness number shall be 35 ± 5. The gasket shall be cemented to the flange part of the top of the case, using adhesive conforming to MMM-A-121 or type II of MMM-A-1617. The entire area of contact between the flange and gasket shall be cemented.

* 3.4.6 Valve, pressure and vacuum relief. The valve shall equalize presssure whenever the pressure differential between the interior and the exterior of the case reaches 1/8 pounds per square inch. The valve shall be capable of withstanding, without degradation of performance attributes, sand and dust particles as specified in MIL-V-27166. The valve housing shall be installed in the case with an airtight seal.

* 3.4.7 Document container. The document container shall be fabricated from 0.090 inch aluminum sheet as specified in 3.4.1. The container shall be 2 inches in diameter by 9-1/2 inches long. The container shall have a twist-on cap captivated by a steel galvanized chain or wire. The container shall be riveted, welded or secured with a nut, washer and sealing gasket to the case. The container opening shall be accessible from the outside of the case.

* 3.4.8 Test plug. A test plug reinforcing plate, fabricated from the aluminum plate, specified in 3.4.1, shall be welded to the inside of the case prior to anodizing.

3.4.9 Clamps. The clamps shall be steel, zinc plated and shall provide a pull down pressure to seal the gasket of the case. The clamps shall be provided with a spring loaded latch finger and latch strike. The spring shall be provided with a takeup to compensate for the set in the gasket. The wing of the latch finger nut shall be hinged to lie flat against the side of the case when locked and shall be actuated by fingertip pressure. A 180 degree turn of the nut shall open or close the latch. After anodizing the case the clamps shall be riveted.

3.4.10 Headrest. One rubber headrest shall be provided in the transfer case. The overall dimensions of the headrest shall be approximately 6-1/4 by 5-1/4 by 4-1/2 inches.

3.5 Performance.

3.5.1 Airtightness. The completely assembled transfer case, when closed and the clamp locks engaged, shall be capable of holding an internal pressure of 1/8 psi gage for 15 minutes with no pressure drop when tested as specified in 4.5.1, and shall be capable of holding a vacuum of 1/8 psi gage for 15 minutes with no drop in vacuum when tested as specified in 4.5.2.

3.5.2 Structural stability. When the pressure of vacuum within the transfer case is raised to 1/4 psi gage plus or minus 10 percent, there shall be no indication of failure or permanent deformation to any part of the transfer case when tested as specified in 4.5.1 and 4.5.2, respectively.

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3.5.3 Stacking. The completely assembled transfer cases shall be capable of being stacked three high to form a stable tier. When tested as specified in 4.5.3, there shall be no rocking of the tier greater than 1/2 inch or other indications of unstableness.

3.6 Finish.

3.6.1 Aluminum. Unless otherwise specified (see 6.2), aluminum components shall be anodized in accordance with type I or II, class 1 of MIL-A-8625.

* 3.6.2 Steel. Steel components, except D-rings and corrosion-resisting steel components, shall be plated in accordance with ASTM B 633.

3.7 Marking.

3.7.1 Identification. The contractor's name, contract number, date of manufacture and letters "US" shall be legibly stamped or stenciled with white waterproof ink or painted on the inside surface of the foot end of the top unit in characters not less than 1/4 inch high.

3.7.2 Special. The following wording shall be placed on the case in clear, legible characters, stamped or stenciled with white, waterproof paint or ink, in the approximate locations shown on the drawing:

The word "HEAD" in characters not less than 1 inch high.
The words "VENT TEST PLUG BEFORE REMOVING COVER" with a leader pointing to the test plug, in characters not less than 1/2 inch high.
The words "PRESSURE RELIEF" in characters not less than 1/4 inch high.

* 3.8 Workmanship. The units shall be free from fractures, splits, punctures, tears, dents, creases, deteriorations or malformations. There shall be no sharp edges, burrs or slivers. The finish shall be free of blisters, peel, chipped areas or areas of no film. The finish shall be free from smudges, poor touch-up, scale, scratches, stain, imprint or embedded foreign matter. Threaded fasteners shall not be broken, stripped or loose. Holes for rivets shall be accurately punched or drilled and any burrs that would prevent proper seating of the rivet be removed. Rivets shall completely fill the holes, with full size heads neatly finished and in full contact with the surface of the riveted members and shall not be loose. The case shall conform to the quality of product established by this specification.

3.8.1 Welders. The welders on the case shall be certified and have passed ASME Welding Qualification Tests (see 2.2).

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3.8.2 Welding. The surfaces of parts to be welded shall be free from oxide, scale, paint, grease, and other foreign matter. Welds shall be continuous, sound, smooth, and free from porosity, cracks, incomplete fusion, and deformation of material. All scale and flux deposits (when flux is used) shall be removed from the finished welds, and rough edges and projections shall be dressed smooth.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

* 4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

* 4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

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* 4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4, and tested for the characteristics specified in 4.4.5.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

* 4.4.1.1 Component and material certification. A certificate of compliance may be acceptable as evidence that the pressure and vacuum relief valve conforms to the requirement specified in 3.4.6.

* 4.4.2 In-process inspection. Inspection of subassemblies shall be made to ascertain that construction details which cannot be examined in the finished product are in accordance with specified requirements. The Government reserves the right to exclude from consideration for acceptance, any material or service for which in-process inspection has indicated nonconformance.

* 4.4.3 End item visual examination. The end items shall be examined for the defects listed in table I. The lot size shall be expressed in units of cases. The sample unit shall be one case. The inspection level shall be II (see 6.4).

TABLE I. End item visual defects

Examine	Defect	Classification Major	Minor
Finish	Not as specified	101	
	Blistered, peeled, chipped or area of no film		201
	Smudges, poor touch-up, scale, scratches, stain, imprint or embedded foreign matter		202
Design, construction and workmanship, general (applicable to all components and assemblies)	Any component or part missing or not specified type or size	102	
	Fractured, split, punctured, torn, dented, creased, deteriorated or malformed	103	
	Sharp burr or edge, or sliver	104	
	Gasket not securely cemented to the flange of the case; area of no adhesion	105	
	Runners not one piece construction	106	

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Design, construction and workmanship general (applicable to all components and assemblies) (cont'd)	Document container not accessible from outside the case, cap not attached to the container	107	
	Test plug not attached to the case		203
Welding	Missing, incomplete, burn holes, cracked, fractured, not fused	108	
	Undercut, not smooth and uniform or scale or flux deposits not removed		204
Bolts, nuts, screws, studs and other types of threaded fasteners	Broken, stripped, fractured or not tight	109	
Rivets	Broken, not peened or insufficiently peened or loose		205
Marking	Missing, incomplete, not legible, not specified type or size, not in proper location, not correct color, not stamped or stenciled		206

* 4.4.4 End item dimensional examination. The end items shall be examined for conformance to the dimensions specified on drawing 5-13-2106. Only those dimensions that can be evaluated without damaging or disassembling the end items shall be examined. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of cases. The sample unit shall be one case. The inspection level shall be S-2 (see 6.4).

* 4.4.5 End item testing. The end items shall be tested as specified in 4.5.1 and 4.5.2. Failure to pass any test shall be cause to reject the affected end item. In addition, the end items shall be tested as specified in 4.5.3, using an inspection level of S-2 (see 6.4).

* 4.4.6 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 (see 6.4).

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Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of container flap, improper taping, loose strapping or inadequate stapling Bulged or distorted container
Content	Number per container is more or less than required

4.5 Methods of inspection.

4.5.1 Pressure test. With the valve removed and the hole plugged, the test plug of the sealed transfer case shall be removed and an air inlet tube equipped with a pressure gage and shutoff cock shall be securely screwed into the test plug hole. Air shall be admitted to the interior of the case until the gage registers 1/8 psi and then the shutoff cock shall be closed. After 15 minutes, the gage reading shall be evaluated to determine compliance with the no pressure drop in internal pressure requirements as specified in 3.5.1. The pressure shall be raised to 1/4 psi gage plus or minus 10 percent and the shutoff cock shall be closed. After 15 minutes the transfer case shall be inspected to determine compliance with the requirements specified in 3.5.2.

4.5.2 Vacuum test. With the relief valve removed and the hole plugged, the test plug of a sealed transfer case shall be removed and a tube equipped with a vacuum gage and shutoff cock shall be securely screwed into the test plug hole. Air shall be drawn from the interior of the transfer case until the gage registers a vacuum of 1/8 psi gage and then the shutoff cock shall be closed. After 15 minutes, the gage reading shall be evaluated to determine compliance with the no drop in vacuum requirements specified in 3.5.1.

* 4.5.3 Stacking test. Three completely assembled and sealed transfer cases shall be stacked three high. The tier shall be visually and manually examined to determine compliance with 3.5.3. In addition, the top case shall be manually rocked by pushing vertically downward and alternatively at diagonally opposite corners to determine compliance with 3.5.3. The force applied shall be adequate to cause rocking of the cases in the event that one or more of the cases is warped or otherwise not flat. The amount of rocking shall be determined with the use of a scale which is secured in position in such a manner to assure an accurate reading.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A.

5.1.1 Level A preservation. The surfaces of the gasket shall be liberally coated with talcum powder conforming to U-T-30 to prevent adhesion to the contacting surfaces. The headrest and tapes shall be placed inside the bottom of the case and immobilized to prevent movement while in transit. The top of the case shall be secured in the closed position with the clamps. The wing nuts on the clamps shall be wire-tied to prevent opening while in transit.

5.2 Packing. Packing shall be level A.

* 5.2.1 Level A packing. Each transfer case, preserved as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Shipments shall be in accordance with Uniform Freight Classification or National Motor Freight Classification, as applicable.

* 5.3 Marking. In addition to any special marking required by the contract or purchase order, shipments shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable. Shipping information shall be marked on tags only, and not on the case.

6. NOTES

* (This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The transfer cases are used to transfer human remains from overseas stations to the US by air transport.

* 6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required (see 3.1, 4.3 and 6.3).
d. When an anodized finish is not required (see 3.6.1).
e. Acceptance criteria required (see 6.4).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

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* 6.4 Acceptance criteria. The acceptance criteria below is recommended for use. The acceptance criteria as specified in the contract or purchase order shall be binding. Unless otherwise specified, the following acceptance criteria are in accordance with MIL-STD-105.

* 6.4.1 For end item visual examination. An acceptable quality level (AQL), expressed in terms of defects per hundred units, of 4.0 for major defects and 6.5 for total (major and minor combined) defects is recommended.

* 6.4.2 For end item dimensional examination. An AQL, expressed in terms of defects per hundred units, of 4.0 is recommended.

* 6.4.3 For end item testing. An AQL, expressed in terms of defects per hundred units, of 4.0 is recommended.

6.4.4 For packaging examination. An AQL, expressed in terms of defects per hundred units, of 2.5 is recommended.

* 6.5 Subject term (key word) listing.

Air transportation
Casket
Cemeterial and mortuary equipment
Transporting

* 6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, irrespective of the marginal notations and relationship to the previous issue.

Custodians:

Army - GL
Navy - NU
Air Force - 99

Preparing activity:

Army - GL

(Project 9930-1003)

Review activities:

Army - MD
DLA - GS

User activities:

Navy - MS
Air Force - 50

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